

Approximations and Errors in Computation

1. Introduction

- In numerical computation, we solve mathematical problems using a finite number of arithmetic operations.
- Since we cannot represent most real numbers exactly in a computer, the results obtained are approximate.
- The difference between the exact (true) value and the computed (approximate) value is called an error.

Numerical methods → Approximate solution with controlled error.

2. Exact Numbers vs Approximate Numbers

- Numbers like 2, -3, 0, $\frac{1}{2}$, $\frac{3}{4}$ are exact in finite arithmetic.
- Numbers like π , $\sqrt{2}$, e etc. cannot be represented exactly. Their computer representation is approximate.

Example:

$$\pi \approx 3.14159265 \text{ (approximate)}$$

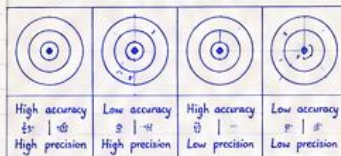
3. Sources of Errors

Errors in computation arise due to:

- Inherent Errors (Data Errors)**
 - Occur due to imperfection in measurement of data.
 - Example: measuring length with a scale.
- Truncation Errors**
 - Occur when an infinite process is replaced by a finite process.
 - Example: using first n terms of a series.
- Round-off Errors**
 - Occur due to finite number of digits used to represent numbers.
 - Example: 7.26923 rounded to 7.269

4. Accuracy and Precision

- Accuracy:** closeness of an approximate value to the true value.
- Precision:** closeness of repeated approximate values to each other.



5. Significant Digits (Figures)

- All non-zero digits are significant.
- Zeros between non-zero digits are significant.
- Leading zeros are not significant.
- Trailing zeros may or may not be significant depending on presence of decimal point.

Examples:

- 0.00345 → 3 significant digits
- 2.0600 → 5 significant digits
- 2001 → 4 significant digits
- 0.0200 → 3 significant digits
- 100.0 → 4 significant digits

6. Absolute Error

Let x be the true value and $\bar{x}$ be the approximate value.

$$\text{Absolute Error, } E_a = |x - \bar{x}|$$

7. Relative Error

$$\text{Relative Error, } E_r = \frac{|x - \bar{x}|}{|x|}$$

8. Percentage Relative Error

$$E_p = E_r \times 100\% = \left| \frac{x - \bar{x}}{x} \right| \times 100\%$$

9. True Error and Approximate Error

- True Error = $x - \bar{x}$ (cannot be found since true value x is usually unknown)
- Approximate Error = estimated error in approximation.

10. Error Propagation

When a result depends on computed quantities, errors in data affect the result.

(i) Addition and Subtraction

$$\text{If } z = x \pm y$$

Then absolute error,

$$E_z = E_x + E_y$$

(ii) Multiplication and Division

$$\text{If } z = x \times y \text{ or } z = \frac{x}{y}$$

Then relative error,

$$\frac{E_z}{|z|} = \frac{E_x}{|x|} + \frac{E_y}{|y|}$$

(iii) General rule:

$$\text{For } z = x^a y^b,$$

$$\frac{E_z}{|z|} = |a| \frac{E_x}{|x|} + |b| \frac{E_y}{|y|}$$

11. Rounding and Chopping

- Rounding:** If the first neglected digit is < 5 , drop it and following digits. If it is ≥ 5 , add 1 to the last retained digit and drop the rest.
Example: 2.34567 rounded to 3 decimal places → 2.346

- Chopping (Truncation):** Simply drop the digits beyond the required place.
Example: 2.34567 chopped to 3 decimal places → 2.345

Note: Rounding gives smaller error than chopping.

12. Floating Point Representation (Brief)

- Computers store real numbers in floating point form:
 $N = \pm m \times b^e$
where
m = mantissa (significand)
b = base (usually 2 in computers)
e = exponent (integer)
- Limited digits in mantissa cause round-off errors.

13. Numerical Stability

- A numerical method is stable if small errors in data produce only small errors in the result.
- An unstable method magnifies the errors.

14. Well-conditioned and Ill-conditioned Problems

- Well-conditioned problem:** Small changes in input → small changes in output.
- Ill-conditioned problem:** Small changes in input → large changes in output.

Example:

$$\text{Solving } ax = b$$

If small change in b causes large change in x, the problem is ill-conditioned.

Good numerical methods should be accurate, stable and well-conditioned.

15. Solved Numerical Examples

Example 1:

True value $x = 25.3421$
Approximate value $\bar{x} = 25.34$
Find absolute error, relative error and percentage relative error.

Solution:

$$E_a = |25.3421 - 25.34| = 0.0021$$

$$E_r = \frac{0.0021}{25.3421} = 0.0000828$$

$$E_p = 0.0000828 \times 100\% = 0.00828\%$$

Example 2:

$x = 47.583$, $y = 12.36$
Absolute errors: $E_x = 0.005$, $E_y = 0.02$
Find absolute error in $z = x - y$

Solution:

$$E_z = E_x + E_y = 0.005 + 0.02 = 0.025$$

Example 3:

Find relative error in $z = \frac{xy}{w}$
where $x = 10 \pm 0.01$, $y = 5 \pm 0.02$,
 $w = 2 \pm 0.005$

Solution:

$$\frac{E_z}{|z|} = \frac{E_x}{|x|} + \frac{E_y}{|y|} + \frac{E_w}{|w|}$$

16. Short Notes for University Exams

- ★ In computation we work with approximate numbers.
- ★ Errors are unavoidable but must be minimized and controlled.
- ★ Main sources of errors are inherent, truncation and round-off errors.
- ★ Accuracy → closeness to true value.
- ★ Precision → closeness among repeated values.
- ★ Smaller the error, higher the accuracy.
- ★ Significant figures indicate the precision of a number.
- ★ Rounding is better than chopping.
- ★ Numerical stability is essential for reliable results.
- ★ Well-conditioned problems give stable solutions.
- ★ Always report results with proper number of significant digits.

17. Formula Summary

- Absolute error: $E_a = |x - \bar{x}|$
- Relative error: $E_r = \frac{|x - \bar{x}|}{|x|}$
- Percentage relative error: $E_p = \left| \frac{x - \bar{x}}{x} \right| \times 100\%$
- Addition/Subtraction: $E_z = E_x + E_y$
- Multiplication/Division: $\frac{E_z}{|z|} = \frac{E_x}{|x|} + \frac{E_y}{|y|}$
- General rule ($z = x^a y^b$): $\frac{E_z}{|z|} = |a| \frac{E_x}{|x|} + |b| \frac{E_y}{|y|}$
- Floating point form: $N = \pm m \times b^e$

18. Frequently Asked Viva Questions

- What is an error in computation?
- Differentiate between accuracy and precision.
- What are the sources of errors?
- Define absolute, relative and percentage relative error.
- What is the difference between rounding and chopping?
- What is machine epsilon?
- What is numerical stability?
- Define well-conditioned and ill-conditioned problems.
- Why are significant digits important?
- State the general rule for error propagation.

Always try to minimize errors and check the reasonableness of results.

19. Practice Problems with Answers

- The true value of a quantity is 18.7265. It is computed as 18.72. Find absolute error, relative error and percentage relative error.
Ans: $E_a = 0.0065$, $E_r = 0.000347$, $E_p = 0.0347\%$
- If $x = 23.4 \pm 0.02$ and $y = 16.8 \pm 0.03$, find absolute error in (i) $x + y$ (ii) $x - y$ (iii) xy (iv) x/y .
Ans: (i) 0.05 (ii) 0.05 (iii) 0.996 (iv) 0.000925
- Find the number of significant digits in (i) 0.00236 (ii) 7.8000 (iii) 100.50
Ans: (i) 3 (ii) 5 (iii) 5

- If $z = \frac{x^2 y}{w^3}$ where $x = 10 \pm 0.1$, $y = 5 \pm 0.05$, $w = 2 \pm 0.02$, find relative error in z.
Ans: $\frac{E_z}{|z|} = 2 \frac{0.1}{10} + \frac{0.05}{5} + 3 \frac{0.02}{2} = 0.02 + 0.01 + 0.03 = 0.06 = 6\%$
- The value of π is taken as 3.1416. Find the absolute error in area of circle of radius 5. (Use $A = \pi r^2$ and $\pi = 3.14159265$)
Ans: $A_{\text{true}} = 3.14159265 \times 25 = 78.53981625$
 $A_{\text{approx}} = 3.1416 \times 25 = 78.5400$
 $E_a = |78.5400 - 78.53981625| = 0.00018375$

★ End of Notes ★